

IoT DEVICE HEALTH MONITORING Over MQTT

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IoT devices are widely used to monitor and control physical systems such as lights, fans, motors, and industrial equipment. This system extends that capability by monitoring the health and operational status of the IoT device itself using the ESP32-C3 Super Mini. The ESP32-C3 continuously monitors key parameters, including CPU usage, RAM utilisation, Wi-Fi signal strength (RSSI), chip temperature, flash memory usage, reset reason, device uptime, and Wi-Fi channel. These telemetry values are periodically published to a public MQTT broker for real-time visualisation on a smartphone dashboard. Bidirectional MQTT communication also enables remote control of the onboard LED and software resets. The system provides a practical,

low-cost solution for remote diagnostics, preventive maintenance, and performance monitoring of embedded IoT devices.

As IoT deployments continue to grow, remote health monitoring has become increasingly important. Continuous telemetry helps identify issues such as memory leaks, poor network connectivity, overheating, and unexpected resets before they result in device failure. This improves system reliability while reducing maintenance effort.

With built-in Wi-Fi, low power consumption, and adequate processing capability, the ESP32-C3 Super Mini is well-suited for a compact health monitoring node. Combined with the lightweight MQTT

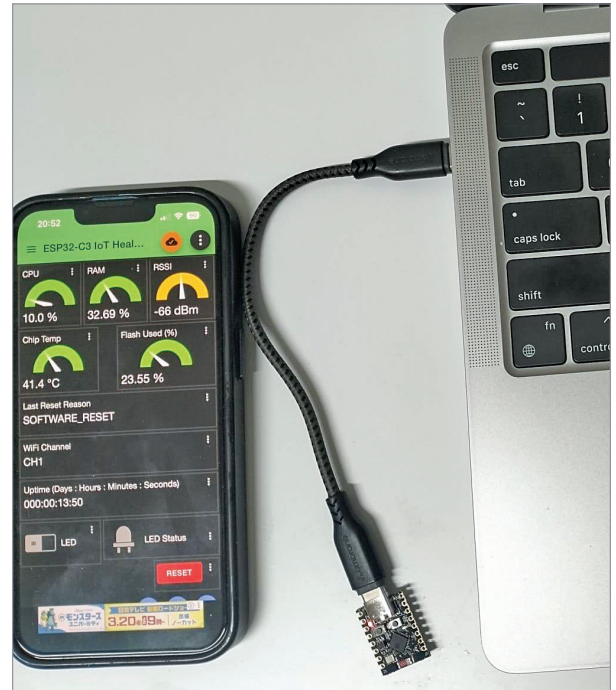


Fig. 1: Authors' prototype setup for IoT device health monitoring

```
const char* ssid = "Galaxy_5GMW_4325";  
const char* password = "xghm1846";  
  
const char* telemetry_topic = "esp32c3/telemetry";  
const char* led_control_topic = "esp32c3/led/control";  
const char* led_status_topic = "esp32c3/led/status";  
const char* esp32c3_reset_topic = "esp32c3/reset";  
const char* uptime_topic = "esp32c3/uptime";
```

Fig. 2: Wi-Fi credentials and MQTT topic configuration in the ESP32-C3 source code

TABLE 1 BILL OF MATERIALS	
Component	Quantity
ESP32-C3 super mini	1
USB Type-C cable	1
Breadboard (optional)	1
Jumper wires	As required
LED (onboard LED may be used)	1
Wi-Fi-enabled smartphone	1
Laptop/PC for programming	1

TABLE 2 MQTT TOPIC STRUCTURE			
Message Description	MQTT Topic	Published By	Subscribed By
Telemetry	esp32c3/telemetry	ESP32-C3 super mini	IoT MQTT panel app
LED control	Esp32c3/led/control	IoT MQTT panel app	ESP32-C3 super mini
LED status	Esp32c3/led/status	ESP32-C3 super mini	IoT MQTT panel app
Remote reset	Esp32c3/reset	IoT MQTT panel app	ESP32-C3 super mini
Formatted uptime	Esp32c3/uptime	ESP32-C3 super mini	IoT MQTT panel app

EFY note. Since this system uses a public MQTT broker, MQTT topic name conflicts may occur. To avoid collisions, append initials, a system identifier, or another unique suffix to each MQTT topic.

protocol, it provides an efficient and scalable solution for real-time IoT device monitoring and remote management. Fig. 1 shows the ESP32-C3 IoT device health monitoring system, while the components required to build the system are listed in Table 1.

MQTT broker configuration

This system uses the Switchsys public MQTT broker, a free MQTT